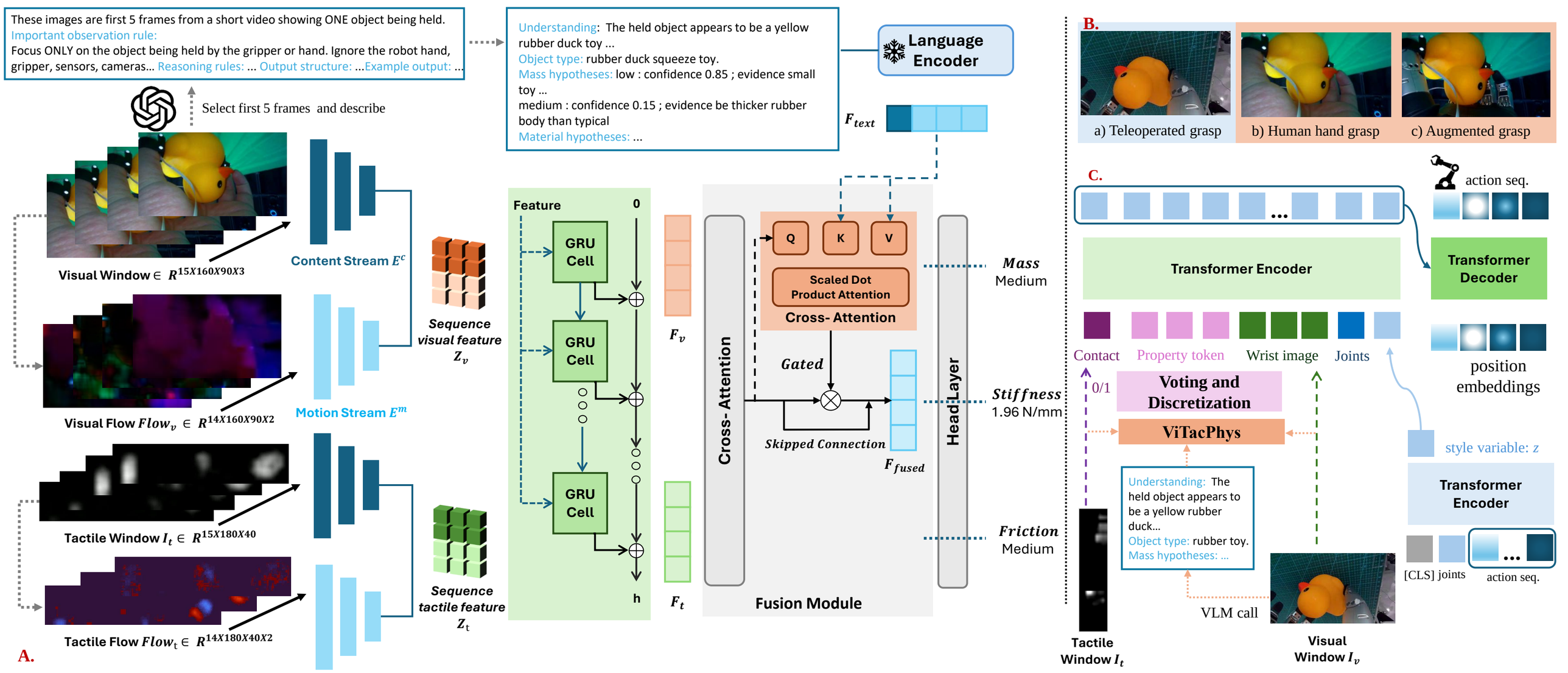

These images are first 5 frames from a short video showing ONE object being held.
Important observation rule:
Focus ONLY on the object being held by the gripper or hand. Ignore the robot hand, gripper, sensors, cameras... Reasoning rules: ... Output structure: ...Example output: ...
Select first 5 frames and describe
Understanding: The held object appears to be a yellow rubber duck toy ...
Object type: rubber duck squeeze toy.
Mass hypotheses: low : confidence 0.85 ; evidence small toy ...
medium : confidence 0.15 ; evidence be thicker rubber body than typical
Material hypotheses: ...
Language Encoder
F_{text}
Visual Window ∈ $R^{15X160X90X3}$
Content Stream E^c
Visual Flow $Flow_v$ ∈ $R^{14X160X90X2}$
Motion Stream E^m
Sequence visual feature Z_v
Tactile Window I_t ∈ $R^{15X180X40}$
Tactile Flow $Flow_t$ ∈ $R^{14X180X40X2}$
Sequence tactile feature Z_t
A.
Feature
0
GRU Cell
GRU Cell
GRU Cell
h
F_v
F_t
Cross- Attention
Q
K
V
Scaled Dot Product Attention
Cross- Attention
Gated
Skipped Connection
F_{fused}
Fusion Module
HeadLayer
Mass
Medium
Stiffness
1.96 N/mm
Friction
Medium
B.
a) Teleoperated grasp
b) Human hand grasp
c) Augmented grasp
C.
action seq.
Transformer Encoder
Transformer Decoder
Contact
Property token
Wrist image
Joints
position embeddings
0/1
Voting and Discretization
ViTacPhys
style variable: z
Understanding: The held object appears to be a yellow rubber duck...
Object type: rubber toy.
Mass hypotheses: ...
Transformer Encoder
[CLS] joints
action seq.
VLM call
Tactile Window I_t
Visual Window I_v